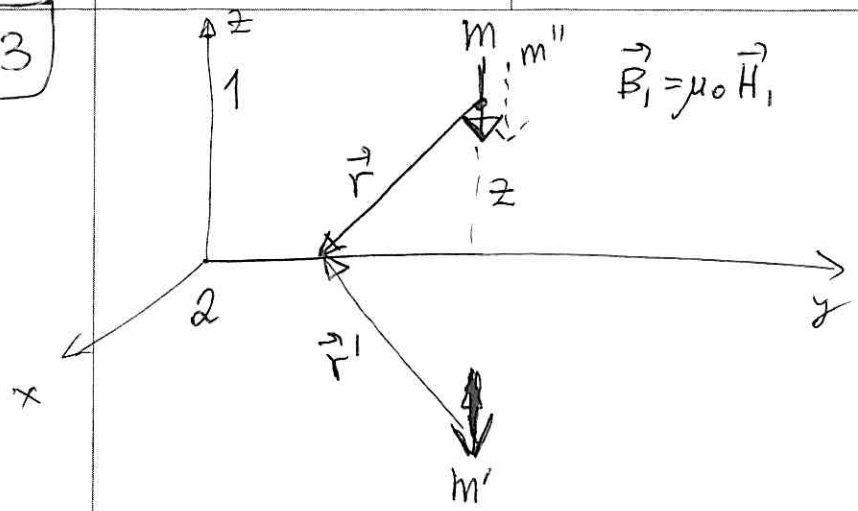
P3



$\vec{B}_1 = \mu_0 \vec{H}_1$

$\vec{B}_2 = \mu \vec{H}_2$

$\vec{\nabla} \times \vec{H} = 0$ here $\Rightarrow$ we can introduce

$$\vec{H} = -\vec{\nabla}\phi_M$$

so that $\nabla^2 \phi_M = 0$.

$\Downarrow$

Can try <u>image construction</u>:

- field in region 1 ($z>0$) is due to dipole $\vec{m}$ and image dipole $\vec{m}' = (-\alpha)\vec{m}$;
- field in region 2 is due to image dipole $\vec{m}'' = \beta \vec{m}$ where β is constant.

- Need to satisfy boundary conditions:

$$H_{1t} = H_{2t}, \quad B_{1\perp} = B_{2\perp} \quad \text{for all } (x, y, z=0)$$

Then in region 1, at $z = +0$, we have (use $\vec{m}' = -\alpha \vec{m}$)

$$\vec{B}_1 = \underbrace{\frac{\mu_0}{4\pi} \frac{3(\vec{m}\cdot\hat{r})\hat{r} - \vec{m}}{r^3}}_{\text{field due to } \vec{m}} + \underbrace{\frac{\mu_0}{4\pi}(-\alpha) \frac{3(\vec{m}\cdot\hat{r}')\hat{r}' - \vec{m}}{(r')^3}}_{\text{field due to } \vec{m}'}$$

$$\vec{B}_2 = \frac{\mu}{4\pi}\beta \frac{3(\vec{m}\cdot\hat{r})\hat{r} - \vec{m}}{r^3}$$

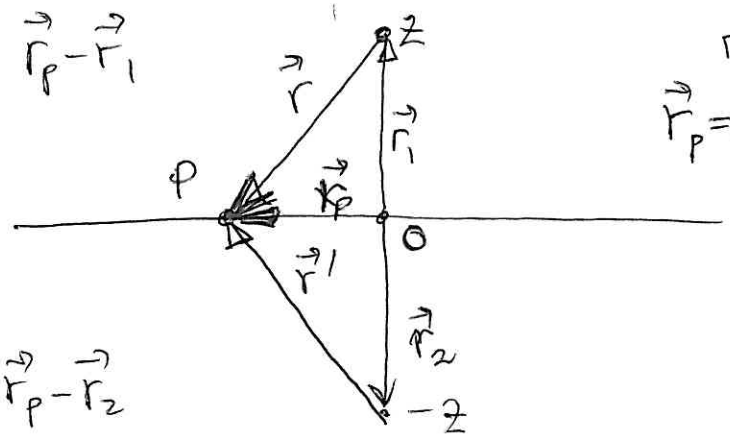
= field due to $\vec{m}'' = \beta \vec{m}$. in medium with permeability μ.

(use $\vec{m}'' = \beta \vec{m}$)

Dipole $\vec{m} = (0, 0, -m)$ at point $\vec{r}_1 = (0, 0, z)$

We assume $\vec{m}' = \alpha \vec{m}$ in medium 2, at the point $\vec{r}_2 = (0, 0, -z)$.

$\vec{r} = \vec{r}_p - \vec{r}_1$



Then point P at the interface has coordinates $\vec{r}_p = (x, y, 0)$.

$\vec{r}' = \vec{r}_p - \vec{r}_2$

With $\vec{m}$ pointing towards the interface we have

$$\vec{m} \cdot \vec{r} = -m\hat{z} \cdot (\vec{r}_p - \vec{r}_1) = mz$$

$$\vec{m} \cdot \vec{r}' = -m\hat{z} \cdot (\vec{r}_p - \vec{r}_2) = -mz$$

$$\vec{H}_1 = \frac{1}{4\pi} \frac{3(\vec{m} \cdot \vec{r})\vec{r} - \vec{m} r^2}{r^5} - \frac{\alpha}{4\pi} \frac{3(\vec{m} \cdot \vec{r}')\vec{r}' - \vec{m}(r')^2}{(r')^5}$$

Note that at the interface $\left.\begin{array}{l} r = |\vec{r}| = \sqrt{x^2+y^2+z^2} \\ r' = |\vec{r}'| = \sqrt{x^2+y^2+z^2} \end{array}\right\} r = r'.$

$$\vec{H}_2 = \frac{\beta}{4\pi} \frac{(3\vec{m} \cdot \vec{r})\vec{r} - \vec{m} r^2}{r^5}$$

$\Rightarrow H_{1t} = H_{2t}$ implies $(\vec{m} \cdot \vec{r})\vec{r} - \alpha(\vec{m} \cdot \vec{r}')\vec{r}' = \beta(\vec{m} \cdot \vec{r})\vec{r}$

$$\Rightarrow \quad (mz)\,\vec{r}_p + \alpha(-mz)\,\vec{r}_p = \beta\,(mz)\,\vec{r}_p$$

$$\Rightarrow \quad 1+\alpha = \beta \quad (1)$$

Then

$$B_{1z} = B_{1z} = \frac{\mu_0}{4\pi}\,\frac{3(\vec{m}\cdot\vec{r})(-z\hat{z}) - (-m\hat{z})\,r^2}{r^5}$$

$$+\frac{\mu_0\alpha}{4\pi}\,\frac{3(\vec{m}\cdot\vec{r}\,')(z\hat{z}) - (-m\hat{z})\,r'^2}{(r')^5}$$

$$B_{2z} = \frac{\mu\beta}{4\pi}\,\frac{3(\vec{m}\cdot\vec{r})(-z\hat{z}) - (-m\hat{z})\,r^2}{r^5}$$

$$\Rightarrow \quad \mu_0(-3mz^2+mr^2) + \mu_0\alpha\,(-3mz^2+mr^2) =$$

$$= \mu\beta\,(-3mz^2+mr^2)$$

$$\Rightarrow 1+\alpha = \frac{\mu}{\mu_0}\beta \quad (2)$$

$$\Rightarrow 2 = \left(1+\frac{\mu}{\mu_0}\right)\beta = \frac{\mu_0+\mu}{\mu_0}\beta \quad , \beta = \frac{2\mu_0}{\mu+\mu_0}$$

(1) $\downarrow \vec{m}$

(2) $\downarrow \vec{m}\,'$

$$\alpha = \beta - 1 = \frac{2\mu_0}{\mu+\mu_0} - 1 = \frac{\mu_0-\mu}{\mu+\mu_0}$$

$$\vec{m}\,' = -\alpha\,\vec{m} = \frac{\mu-\mu_0}{\mu+\mu_0}\,\vec{m} \quad (\text{parallel to } \vec{m}!)$$

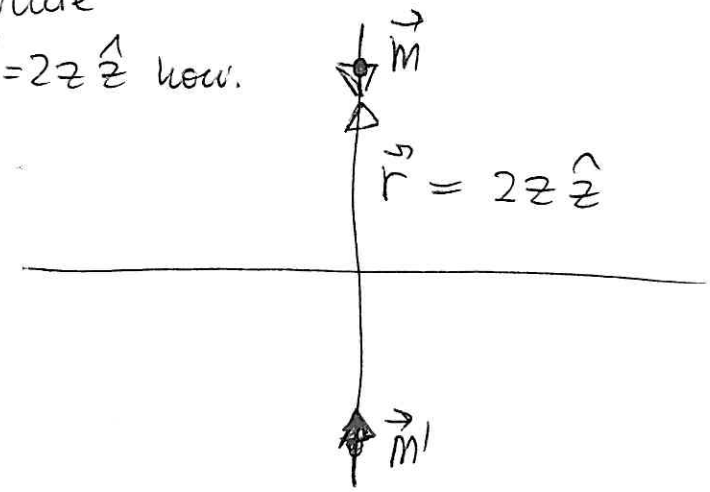
Force acting on $\vec{m}$ is due to its image $\vec{m}' = -\alpha \vec{m}$

$$\vec{F} = \vec{\nabla}\left(\vec{m} \cdot \vec{B}'\right)$$

$$\vec{B}' = \frac{\mu_0}{4\pi} \frac{3\left(\vec{m}' \cdot \vec{r}\right)\vec{r} - \vec{m}' r^2}{r^5}$$

where

$\vec{r} = 2z\hat{z}$ here.



$$\vec{m} \cdot \vec{r} = -m(2z)$$

$$\vec{B}' = -\frac{\mu_0 \alpha}{4\pi} \frac{3\left(\vec{m} \cdot \vec{r}\right)\vec{r} - \vec{m} r^2}{r^5} =$$

$$= -\frac{\mu_0 \alpha}{4\pi} \frac{3(-2zm)(2z\hat{z}) - (-m\hat{z})(2z)^2}{(2z)^5}$$

$$= \frac{\mu_0 \alpha}{4\pi} \frac{3m\hat{z} - m\hat{z}}{(2z)^3} = \frac{\mu_0 \alpha}{4\pi} \frac{m\hat{z}}{4z^3}$$

$$\vec{B}' \cdot \vec{m} = -\frac{\mu_0 \alpha}{4\pi} \frac{m^2}{4z^3}$$

$$\Rightarrow \vec{F} = \frac{3\mu_0 \alpha m^2}{16\pi z^4}\hat{z} = -\frac{3\mu_0(\mu - \mu_0)}{16\pi z^4} m^2 \hat{z}$$

The dipole is attracted to the interface.

Another way: think of $\vec{m}$ as being produced by a current loop — the current $\vec{J}(r)$ is in x-y plane parallel to the interface

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μ

Problem 5.17 of Jacuson's text (HW 12) shows that image current $\vec{J}' = \frac{\mu_r - 1}{\mu_r + 1}\vec{J} = \frac{\mu - \mu_0}{\mu + \mu_0}\vec{J}$

(for x, y components of the current $\vec{J}, \vec{J}'$)

Since $\vec{m} = \frac{1}{2}\int \vec{r} \times \vec{J}(r)\, d^3r$

this means that $\vec{m}' = \frac{1}{2}\int \vec{r}' \times \vec{J}'(\vec{r}')\, d^3\vec{r}'$

$= \frac{\mu - \mu_0}{\mu + \mu_0}\vec{m}$ just as we found on page 3.

Thus $\vec{J}'$ is circling in the same way as $\vec{J}$.

Since parallel currents ~~[illegible]~~ are attracted to each other, $\vec{m}$ is attracted to $\vec{m}'$, that is, attracted to the medium with higher μ.